

Notes Power Model D-MIMO

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I. CENTRAL MIMO POWER MODEL

We adopt the parametric power model for upper mid-band (FR3) base stations of [1]. The base station (BS) is equipped with M_{ant} antennas, an equal number of power amplifiers (PAs, used in the downlink) and low-noise amplifiers (LNAs, used in the uplink), and M_{RF} radio-frequency (RF) chains. It serves K single-antenna users over a bandwidth B at carrier frequency f_c using OFDM and space-division multiple access. Setting $M_{\text{RF}} = M_{\text{ant}}$ selects fully-digital beamforming; $M_{\text{RF}} < M_{\text{ant}}$ selects hybrid partially-connected beamforming with $M_{\text{ant}}/M_{\text{RF}}$ phase shifters per RF chain.

A. Model structure

The total consumed power is split into three hardware blocks; digital signal processing, analog signal processing, and the PAs, each divided by its supply-and-cooling efficiency $\eta_{c,\text{sc}} \in (0, 1]$:

$$P_{\text{cons}} = \frac{\overline{P_{\text{dig}}}}{\eta_{\text{dig,sc}}} + \frac{\overline{P_{\text{ana}}}}{\eta_{\text{ana,sc}}} + \frac{\overline{P_{\text{PA}}}}{\eta_{\text{PA,sc}}}. \quad (1)$$

Each block is built from the working-mode (active) consumption of its subcomponents, averaged over the operating modes within a frame. We first define this frame averaging (Section I-B) and then reuse it as a function for every component (Sections I-C to I-E).

B. Frame structure and operating-mode averaging

A frame of duration T_f contains a downlink (DL) and an uplink (UL) phase under time division duplexing (TDD), with duration ratios $\tau_{\text{DL}}, \tau_{\text{UL}} \in [0, 1]$ and $\tau_{\text{UL}} = 1 - \tau_{\text{DL}}$. Within each direction $i \in \{\text{DL}, \text{UL}\}$ the time is further subdivided into three operating modes, plus an idle mode when the direction is inactive:

- *reference signalling*, for a time fraction $\tau_i \tau_{i,\text{sig}}$;
- *data transmission/reception*, for a fraction $\tau_i (1 - \tau_{i,\text{sig}}) \frac{N_{a,i}}{N_i}$;
- *micro-sleep* (DTX), for a fraction $\tau_i (1 - \tau_{i,\text{sig}}) \left(1 - \frac{N_{a,i}}{N_i}\right)$;
- *idle*, for a fraction $(1 - \tau_i)$,

where N_i is the number of OFDM symbols (time slots) of direction i and $N_{a,i} \leq N_i$ the subset that are *active* (carrying data). The *active-resource time ratio* $N_{a,i}/N_i$ is what sets the data-versus-micro-sleep split: a slot is either active or not, regardless of how many users share it. with $\tau_{i,\text{sig}} \in [0, 1]$ the signalling fraction of direction i .

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a) *Average physical resource load*: The data-mode weight is more conveniently expressed through the *average physical resource load* $\bar{x}_i$, which we now relate to the time ratio $N_{a,i}/N_i$. Let K_{max} be the maximum number of users that can be served (the number of spatial streams) and Q_{max} the number of available data subcarriers per active OFDM symbol. The instantaneous load of user k in time slot n is the ratio of utilised to available data subcarriers,

$$\ell_i[k, n] = \frac{Q_i[k, n]}{Q_{\text{max}}}, \quad i \in \{\text{DL}, \text{UL}\}, \quad (2)$$

for $k = 1, \dots, K_{\text{max}}$ and $n = 1, \dots, N_i$. Averaging over all K_{max} potential users and N_i slots gives the average physical resource load

$$\bar{x}_i \triangleq \bar{\ell}_i = \frac{1}{N_i} \sum_{n=1}^{N_i} \left(\frac{1}{K_{\text{max}}} \sum_{k=1}^{K_{\text{max}}} \frac{Q_i[k, n]}{Q_{\text{max}}} \right). \quad (3)$$

Under space-division multiple access (no OFDMA), an active user in an active slot is allocated *all* data subcarriers, $Q_i[k, n] = Q_{\text{max}}$. Indexing the K_i active users as $\{1, \dots, K_i\}$ and the $N_{a,i}$ active slots as $\{1, \dots, N_{a,i}\}$, each active-user/active-slot pair contributes 1 to (3) and every other pair 0; the double sum equals $K_i N_{a,i}$ and

$$\bar{x}_i = \frac{N_{a,i}}{N_i} \frac{K_i}{K_{\text{max}}}. \quad (4)$$

b) *Equivalence and modelling assumption*: The duty-cycle split in the frame structure is governed by the active-resource time ratio $N_{a,i}/N_i$, whereas the power model uses the average load $\bar{x}_i$ as the data-mode weight. Comparing (4) with the data fraction $\tau_i (1 - \tau_{i,\text{sig}}) N_{a,i}/N_i$, the two agree,

$$\bar{x}_i = \frac{N_{a,i}}{N_i} \iff \frac{K_i}{K_{\text{max}}} = 1 \iff K_i = K_{\text{max}}, \quad (5)$$

i.e. only when every spatial stream is scheduled. When the maximum number of users is active the average load is simply the ratio of active to total slots; for $K_i < K_{\text{max}}$ the average load (4) is the true active fraction scaled by $K_i/K_{\text{max}} < 1$. The number of active users K_i separately sets the transmit power and the achievable rate, but not whether a slot is active. We use $\bar{x}_i$ as the data-mode weight throughout (and $1 - \bar{x}_i$ as the micro-sleep/DTX fraction); this is therefore *exact* in the full-spatial-load regime $K_i = K_{\text{max}}$ evaluated here and an approximation otherwise.

In micro-sleep and idle a component drops to a fraction of its working consumption, set by the reduction factors $\delta_c^\mu, \delta_c^0 \in [0, 1]$. For a component c with working power P^a during data, P^s during signalling, and base power P^0 scaled in the reduced

modes, the frame-averaged consumption of direction i is the time-weighted sum

$$\begin{aligned} \mathcal{A}_i^c(P^a, P^s, P^0) &\triangleq \bar{x}_i \tau_i (1 - \tau_{i,\text{sig}}) P^a + \tau_i \tau_{i,\text{sig}} P^s \\ &\quad + (1 - \bar{x}_i) \tau_i (1 - \tau_{i,\text{sig}}) \delta_c^\mu P^0 \\ &\quad + (1 - \tau_i) \delta_c^0 P^0. \end{aligned} \quad (6)$$

For the digital and analog blocks the consumption is identical in all working modes, $P^a = P^s = P^0 = P$, and we use the shorthand $\mathcal{A}_i^c(P) \triangleq \mathcal{A}_i^c(P, P, P)$. Only the PA distinguishes the three levels (Section I-C).

Each direction's hardware is averaged over the *whole* frame, and the two directions are accounted for separately, e.g. for the digital block $\bar{P}_{\text{dig}} = \mathcal{A}_{\text{DL}}^{\text{dig}}(P_{\text{dig},\text{DL}}) + \mathcal{A}_{\text{UL}}^{\text{dig}}(P_{\text{dig},\text{UL}})$. The idle term $(1 - \tau_i) \delta_c^0 P^0$ is therefore the reduced consumption of direction i 's chain while that direction is *not* active, i.e. while the opposite direction occupies the frame. For $i = \text{DL}$ this fraction equals τ_{UL} only because the gap-free TDD frame satisfies $\tau_{\text{UL}} = 1 - \tau_{\text{DL}}$; it represents the idle DL chain, not the active UL processing, which is captured by the separate $\mathcal{A}_{\text{UL}}$ term.

Remark 1 (Load-dependent split). *Only the data term in (6) scales with the load $\bar{x}_i$, while the micro-sleep term scales with $(1 - \bar{x}_i)$. Collecting the terms proportional to $\bar{x}_i$ separates each averaged consumption into a load-dependent part and a load-independent part, which is used in the analysis to attribute consumption to the delivered traffic.*

C. Power amplifiers

The PAs transmit only in the DL. With a per-PA output power $P_a = P_{T,\text{DL}}/M_{\text{ant}}$, where $P_{T,\text{DL}}$ is the total DL transmit power, the instantaneous consumption of one PA is modelled as

$$P_{\text{PA}}(p) = \frac{1}{\eta_{\text{PA},\text{max}}} [\xi P_{\text{max}} + (1 - \xi) P_{\text{max}}^{1-\alpha} p^\alpha], \quad (7)$$

where p is the output power, P_{max} the maximum output power (at a fixed back-off from saturation), $\eta_{\text{PA},\text{max}}$ the maximum PA efficiency, $\xi \in [0, 1]$ the static-to-dynamic consumption ratio, and $\alpha \in [0, 1]$ the efficiency exponent. Averaging over the DL frame and summing over the M_{ant} PAs gives

$$\overline{P_{\text{PA}}} = M_{\text{ant}} \mathcal{A}_{\text{DL}}^{\text{PA}}(P_{\text{PA}}(P_a), P_{\text{PA}}(\zeta_{\text{sig}} P_{\text{max}}), P_{\text{PA}}(0)), \quad (8)$$

where $\zeta_{\text{sig}} \in (0, 1]$ is the fraction of P_{max} used during signalling, and the zero-output consumption $P_{\text{PA}}(0)$ is the base level scaled in micro-sleep and idle.

D. Analog signal processing

The analog block comprises, per RF chain, the DACs (DL) or ADCs (UL), RF filters, mixers, M_{PS} phase shifters, and the LNAs (UL); the local oscillator (LO) is shared across chains. With $M_{\text{PS}} = M_{\text{ant}}/M_{\text{RF}}$ for hybrid beamforming ($M_{\text{PS}} = 0$

for fully-digital), the working-mode consumption per direction is

$$P_{\text{ana},\text{DL}} = P_{\text{LO}} + M_{\text{RF}}(2P_{\text{DAC}} + 2P_{\text{filt},\text{RF}} + 2P_{\text{mix}} + M_{\text{PS}}P_{\text{PS}}), \quad (9)$$

$$P_{\text{ana},\text{UL}} = P_{\text{LO}} + M_{\text{RF}}\left(2P_{\text{ADC}} + 2P_{\text{filt},\text{RF}} + 2P_{\text{mix}} + M_{\text{PS}}P_{\text{PS}} + \frac{M_{\text{ant}}}{M_{\text{RF}}}P_{\text{LNA}}\right), \quad (10)$$

with the factor 2 accounting for the in-phase and quadrature branches. The subcomponents are modelled as

$$P_{\text{DAC}} = \Xi_{\text{DAC},1} 2^{b_{\text{DAC}}} + \Xi_{\text{DAC},2} b_{\text{DAC}} f_{\text{DAC}}, \quad (11)$$

$$P_{\text{ADC}} = W_{\text{ADC}} f_{\text{ADC}} 2^{b_{\text{ADC}}}, \quad (12)$$

$$P_{\text{mix}} = \Xi_{\text{mix}} f_c, \quad (13)$$

$$P_{\text{PS}} = \Xi_{\text{PS}} B, \quad (14)$$

$$P_{\text{LNA}} = \Xi_{\text{LNA}} B, \quad (15)$$

where $b_{\text{DAC}}, b_{\text{ADC}}$ are the effective resolutions, $f_{\text{DAC}}, f_{\text{ADC}}$ the converter sampling rates, W_{ADC} the ADC Walden figure-of-merit, and $\Xi_\bullet$ the respective figures-of-merit. The frame-averaged analog consumption is

$$\overline{P_{\text{ana}}} = \mathcal{A}_{\text{DL}}^{\text{ana}}(P_{\text{ana},\text{DL}}) + \mathcal{A}_{\text{UL}}^{\text{ana}}(P_{\text{ana},\text{UL}}) + P_{\text{ana},\text{misc}}. \quad (16)$$

E. Digital signal processing

In the DL the digital block runs the channel encoder, MIMO precoder, IFFT, DPD, and baseband (BB) filter; in the UL the channel decoder, MIMO combiner, FFT, and BB filter. Each subcomponent consumption is a static (signal-independent) term plus a dynamic term equal to its complex giga-operations-per-second (GOPS) Ξ divided by the computational efficiency η (in GOPS/W) of the chosen FPGA/ASIC implementation:

$$P_{\text{enc}} = P_{\text{enc},s} + \frac{f_{s,\text{I}}}{\eta_{\text{enc}}} \frac{14}{24} \frac{R_{\text{DL}}}{\tilde{B}}, \quad (17)$$

$$P_{\text{dec}} = P_{\text{dec},s} + \frac{f_{s,\text{I}}}{\eta_{\text{dec}}} \frac{175}{6} \frac{R_{\text{UL}}}{\tilde{B}}, \quad (18)$$

$$P_{\text{pre}} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,\text{I}}}{\eta_{\text{pre}}} M_{\text{RF}} \Xi_{\text{pre}}, \quad (19)$$

$$P_{\text{comb}} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,\text{I}}}{\eta_{\text{comb}}} M_{\text{RF}} 2K, \quad (20)$$

$$P_{\text{IFFT}} = P_{\text{IFFT},s} + \frac{f_{s,\text{II}}}{\eta_{\text{IFFT}}} M_{\text{RF}} \frac{3}{2} \log_2 Q_{\text{IFFT}}, \quad (21)$$

$$P_{\text{DPD}} = P_{\text{DPD},s} + \frac{f_{s,\text{II}}}{\eta_{\text{DPD}}} M_{\text{RF}} \Xi_{\text{DPD}}, \quad (22)$$

$$P_{\text{filt},\text{BB}} = \bar{M}_{\text{RF}} P_s + \frac{f_{s,\text{II}}}{\eta_{\text{BB}}} M_{\text{RF}} \Xi_{\text{BB}}, \quad (23)$$

where $R_{\text{DL}}, R_{\text{UL}}$ are the delivered DL/UL sum rates, $\tilde{B}$ the effective bandwidth, $\bar{M}_{\text{RF}} = \lceil M_{\text{RF}}/32 \rceil$ the number of FPGAs (one per 32 chains, sharing the precoder/combiner/BB-filter static power P_s), and Q_{IFFT} the IFFT/FFT size. The two sampling rates are the constellation rate $f_{s,\text{I}} = \mu B$ (for encoding/decoding and MIMO processing) and the oversampled rate $f_{s,\text{II}} = Q_{\text{IFFT}} \Delta f$ (for the OFDM, DPD, and BB operations), with $\mu \in [0, 1]$ and subcarrier spacing Δf . For the zero-forcing

MIMO processor exploiting channel reciprocity, the precoder GOPS is

$$\Xi_{\text{pre}} = 2K + \frac{1}{v_{\text{coh}}} \left(\frac{K^3}{3M_{\text{RF}}} + 3K^2 + K \right), \quad (24)$$

with v_{coh} the number of samples per channel coherence block over which the precoder is reused; the combiner only applies the precoding matrix, giving $\Xi_{\text{comb}} = 2K$. The active digital consumption per direction is the sum of its subcomponents,

$$P_{\text{dig,DL}} = P_{\text{enc}} + P_{\text{pre}} + P_{\text{IFFT}} + P_{\text{DPD}} + P_{\text{filt,BB}}, \quad (25)$$

$$P_{\text{dig,UL}} = P_{\text{dec}} + P_{\text{comb}} + P_{\text{IFFT}} + P_{\text{filt,BB}}, \quad (26)$$

and the frame-averaged digital consumption is

$$\overline{P_{\text{dig}}} = \mathcal{A}_{\text{DL}}^{\text{dig}}(P_{\text{dig,DL}}) + \mathcal{A}_{\text{UL}}^{\text{dig}}(P_{\text{dig,UL}}). \quad (27)$$

F. Delivered rate and energy efficiency

The achievable ergodic sum rate of direction i , which drives the encoder/decoder consumption in (17), follows from the per-user SINRs under zero-forcing beamforming as

$$R_i = \bar{x}_i \tau_i (1 - \tau_{i,\text{sig}}) \frac{\tilde{B}}{q_i} \mathbb{E} \left\{ \sum_{\nu=1}^{q_i} \sum_{k=1}^K \log_2(1 + \text{SINR}_{i,k}[\nu]) \right\}, \quad (28)$$

where the pre-log collects the load, duration, signalling-overhead, and effective-bandwidth-per-subcarrier factors. The base-station energy efficiency then follows as the delivered rate per consumed watt,

$$\text{EE} = \frac{R_{\text{DL}} + R_{\text{UL}}}{P_{\text{cons}}} \quad [\text{bit/J}]. \quad (29)$$

REFERENCES

- [1] E. Peschiera, S. Yun, Y. Lee, L. V. der Perre, and F. Rottenberg, "A parametric power model of upper mid-band (fr3) base stations for 6g," in *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 21 476–21 480.